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§ 5. Computation with Units and Reducibility into k Subgroups.

The residue classes $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{G}$ share with the unit $\mathfrak{e}$ the property that the product of $\mathfrak{a}$ and $\mathfrak{b}$ with an arbitrary residue class $\mathfrak{x}$ is defined; we therefore likewise call $\mathfrak{a}$ and $\mathfrak{b}$ units of $\mathfrak{G}$. Indeed, from (11), for $F = M$, there follows

$$0 = M\mathfrak{e} = M\mathfrak{a} + M\mathfrak{b},$$

and hence, by uniqueness, $M\mathfrak{a} = 0$, $M\mathfrak{b} = 0$, and therefore

$$(F + M)\mathfrak{a} = F\mathfrak{a} = \mathfrak{x}\mathfrak{a},$$

where $\mathfrak{x}$ denotes the residue class of F . In place of relation (11), we may therefore write the equivalent class relation

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a} + \mathfrak{x}\mathfrak{b}.$$

In particular, putting $\mathfrak{x} = \mathfrak{a}$ gives

$$\mathfrak{a} = \mathfrak{a}^2 + \mathfrak{a}\mathfrak{b},$$

and hence, by uniqueness, $\mathfrak{a}^2 = \mathfrak{a}$, $\mathfrak{a}\mathfrak{b} = 0$. Similarly one obtains $\mathfrak{b}^2 = \mathfrak{b}$, $\mathfrak{b}\mathfrak{a} = 0$.

Corresponding to the representation of a residue group as a sum of two constituents, we may now also define such a representation as a sum of k constituents. Namely, suppose a unique representation is given

$$F\mathfrak{e} = F\mathfrak{a}_1 + F\mathfrak{a}_2 + \cdots + F\mathfrak{a}_k \quad (\mathfrak{a}_1 \neq 0, \dots, \mathfrak{a}_k \neq 0),$$

where from $0 = G_1\mathfrak{a}_1 + \cdots + G_k\mathfrak{a}_k$ it also follows that $G_i\mathfrak{a}_i = 0$. Then, in particular for $F = M$, one obtains the relation $M\mathfrak{a}_i = 0$, so that the representation may also be written in the form¹

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_k \quad (\mathfrak{a}_1 \neq 0, \dots, \mathfrak{a}_k \neq 0). \quad (12)$$

This representation may also be regarded as having arisen from a successive decomposition into two constituents. If this can be continued without limit, with all $\mathfrak{a}_i \neq 0$, then we speak of a sum of infinitely many constituents.

Putting $\mathfrak{x} = \mathfrak{a}_i$ in (12) gives

$$\mathfrak{a}_i^2 = \mathfrak{a}_i, \quad \mathfrak{a}_i\mathfrak{a}_j = 0 \quad (i \neq j), \quad i, j = 1, \dots, k, \quad (13)$$

with $\mathfrak{a}_i \neq 0$. Conversely, suppose a system of residue classes $\mathfrak{a}_1, \dots, \mathfrak{a}_k$ in $\mathfrak{G}$ is present for which the orthogonality conditions (13) are fulfilled and for which

$$\mathfrak{e} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_k \quad (14)$$

holds. We shall show that this gives a representation of the group as a sum of k constituents. First, from the hypothesis, for every polynomial A_i from $\mathfrak{a}_i$, since $A_i\mathfrak{a}_i = (A_i + M)\mathfrak{a}_i$, it follows that $M\mathfrak{a}_i = 0$; hence $\mathfrak{a}_i$ can be multiplied by every class. Thus (14) implies representation (12), and this representation is unique. For if $0 = \mathfrak{x}_1\mathfrak{a}_1 + \cdots + \mathfrak{x}_k\mathfrak{a}_k$, then right multiplication by $\mathfrak{a}_i$ gives, by (13), $0 = \mathfrak{x}_i\mathfrak{a}_i$, proving uniqueness and hence the assertion. The residue classes $\mathfrak{a}_1, \dots, \mathfrak{a}_k$ occurring in the decomposition of the unit class will be called the units belonging to the decomposition; equivalently they are characterized by the orthogonality relations (13) and condition (14). The later developments will essentially amount to computation with units.

It remains to establish the connection between the additive decomposition of the group into k constituents and the corresponding decomposition of the module. As in the case $k = 2$, a representation of the module as the least common multiple of k relatively prime modules will result, by induction.

Let the decomposition (12) be given, which we write in the form

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{b} + \mathfrak{x}\mathfrak{a}_k, \quad (15)$$

¹For the additive decomposition of residue groups, the commutative and associative laws hold; in contrast to the commutative case, however, $\mathfrak{u}_1 + \mathfrak{u}_2 = \mathfrak{u}_1 + \mathfrak{u}_3$ does not imply $\mathfrak{u}_2 = \mathfrak{u}_3$; cf. example 1 in § 12.

$$\mathfrak{r}\mathfrak{b} = \mathfrak{r}\mathfrak{a}_1 + \cdots + \mathfrak{r}\mathfrak{a}_{k-1}. \quad (16)$$

From (15) it follows by Theorem I that the residue classes $\mathfrak{r}\bar{\mathfrak{b}}$ are isomorphic to the residue group of the module

$$\mathfrak{N} = (\mathfrak{M}, A_k). \quad (17)$$

For their residue classes $\mathfrak{r}\bar{\mathfrak{b}}$, the representation corresponding to (16) is therefore also valid:

$$\mathfrak{r}\bar{\mathfrak{b}} = \mathfrak{r}\bar{\mathfrak{a}}_1 + \cdots + \mathfrak{r}\bar{\mathfrak{a}}_{k-1}. \quad (18)$$

Suppose now it has already been proved that

$$\mathfrak{N} = [\mathfrak{M}_1, \dots, \mathfrak{M}_{k-1}], \quad (19)$$

where

$$\begin{cases} \mathfrak{M}_1 = (\mathfrak{N}, A_2 + \cdots + A_{k-1}), \\ \mathfrak{M}_2 = (\mathfrak{N}, A_1 + A_3 + \cdots + A_{k-1}), \\ \vdots \\ \mathfrak{M}_{k-1} = (\mathfrak{N}, A_1 + \cdots + A_{k-2}). \end{cases} \quad (20)$$

For two summands this assertion agrees with our earlier result. Moreover, the polynomials A_i may here be chosen specifically as polynomials from $\bar{\mathfrak{a}}_i$ which at the same time are contained in $\mathfrak{a}_i$, since by § 4 the polynomials of the latter residue class are contained in the former. Then, by (15),

$$\mathfrak{M} = [\mathfrak{N}, \mathfrak{M}_k], \quad (21)$$

where

$$\mathfrak{M}_k = (\mathfrak{M}, A_1 + \cdots + A_{k-1}).$$

From (19) and (21) one obtains

$$\mathfrak{M} = [[\mathfrak{M}_1, \dots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{M}_1, \dots, \mathfrak{M}_{k-1}, \mathfrak{M}_k]$$

by the definition of least common multiple; and from (17) and (20) one gets

$$\mathfrak{M}_1 = (\mathfrak{M}, A_k, A_2 + \cdots + A_{k-1}).$$

It follows that

$$\mathfrak{M}_1 = (\mathfrak{M}, A_2 + \cdots + A_k),$$

for the latter module contains, by (13), also the polynomial

$$A_k = A_k(A_2 + \cdots + A_k) \pmod{\mathfrak{M}}.$$

Corresponding expressions are obtained for $\mathfrak{M}_2, \dots, \mathfrak{M}_{k-1}$. Furthermore, the modules $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ are relatively prime, since by (14)

$$\frac{1}{k-1} \left((A_2 + \cdots + A_k) + (A_1 + A_3 + \cdots + A_k) + \cdots + (A_1 + \cdots + A_{k-1}) \right) \equiv 1 \pmod{\mathfrak{M}}.$$

They moreover form a totally relatively prime system, that is, the least common multiple of any of them is relatively prime to the least common multiple of the remaining ones.² This follows directly by grouping the summands in formula (12) into two suitable groups.

It has thus been shown that to a decomposition of the residue group into k summands there corresponds a representation of the module as the least common multiple of k modules forming a totally relatively prime system. We now prove the converse theorem, likewise by induction.

Suppose therefore that

$$\mathfrak{M} = [\mathfrak{M}_1, \dots, \mathfrak{M}_k] = [[\mathfrak{M}_1, \dots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{N}, \mathfrak{M}_k],$$

²This is a generalization of a concept introduced by Loewy.

where $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ form a totally relatively prime system. In combination with Theorem I this gives representation (15) for the residue group. The modules $\mathfrak{M}_1, \dots, \mathfrak{M}_{k-1}$ also form a totally relatively prime system. For if, under some division of the modules of this system into two groups, the least common multiple of one group and the least common multiple of the other had a divisor in common, this divisor would remain if the module $\mathfrak{M}_k$ were added to one of the two groups and the least common multiple were then formed.³ For $k = 2$, “totally relatively prime” means the same as “relatively prime”; hence our assertion may be assumed proved up to $k - 1$. Thus for the residue class $\bar{x}$ the unique representation (18) holds, hence, by the isomorphism, also (16), and so the assertion follows.

Theorem II. *A module $\mathfrak{M}$ is representable as the least common multiple of k modules $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ which form a totally relatively prime system if and only if its residue group $\mathfrak{S}$ is reducible into k subgroups $\mathfrak{U}_1, \dots, \mathfrak{U}_k$. With suitable notation, $\mathfrak{U}_i$ is then isomorphic to the residue group of $\mathfrak{M}_i$.*

§ 6. Finiteness Theorems.

In this paragraph it will be shown that every residue group can be represented as a sum of finitely many irreducible constituents. To do this we first transfer Hilbert’s theorem on the module basis.

From now on we use the more special hypotheses formulated in § 2: the multiplication of $\xi_1, \dots, \xi_n$ among themselves is commutative, and the multiplication law of the polynomials has the form

$$\xi_i a = a_i \xi_i + b_i, \quad (a_i \neq 0) \quad (i = 1, \dots, n). \quad (6)$$

It suffices to prove the assertion for modules only. For if $\mathfrak{S}$ is any system of polynomials, we form the module $\mathfrak{M}$ derived from it by adjoining to $\mathfrak{S}$ all polynomials which can be composed from finitely many polynomials $G_1, \dots, G_k$ of $\mathfrak{S}$ in the form $A_1 G_1 + \dots + A_k G_k$. If $F_1, \dots, F_l$ is then a module basis of $\mathfrak{M}$, and if

$$F_i = \sum_j A_{ij} G_j \quad (i = 1, \dots, l),$$

then the finitely many polynomials G_j occurring in these representations form a basis of $\mathfrak{S}$.

We prove the module-basis theorem following Gordan (Göttingen Nachrichten 1899). The proof splits into two parts. The first concerns a system $\mathfrak{P}$ of infinitely many power products and remains essentially unchanged because of the commutativity of $\xi_1, \dots, \xi_n$. The power products

$$P = \xi_1^{a_1} \dots \xi_n^{a_n},$$

each of which is entered into the system only once, are to be ordered by increasing degree and, within the same degree, in some fixed way. We now consider the subsystem $\mathfrak{Q}$ of $\mathfrak{P}$ consisting of all power products $P = Q$ not divisible by any preceding P . Then no Q is divisible by a later Q either, since later ones are either of higher degree or, in the same degree, different from the earlier ones. On the other hand, every P of $\mathfrak{P}$ is divisible by at least one Q . For suppose, to the contrary, that P_m is the first P divisible by no Q . Then P_m cannot itself be a Q , and so it is divisible by at least one preceding P ; since by hypothesis this preceding P is divisible by some Q , P_m itself is divisible by a Q , a contradiction.

We now assert that the number of the Q is finite. Indeed, let

$$Q_0 = \xi_1^{h_1} \dots \xi_n^{h_n}$$

be the first Q . Since an arbitrary $Q_\lambda = \xi_1^{h_{1\lambda}} \dots \xi_n^{h_{n\lambda}}$ is not divisible by Q_0 , for every Q at least one $h_{\rho\lambda} < h_\rho$ must hold; let λ denote the smallest index for which this is the case. We group all Q for which $h_{\rho\lambda}$ has the same fixed value c_λ into one class $K(c_\lambda)$. Since $c_\lambda < h_\lambda$, the number of these classes is finite. But each class contains only finitely many elements. For if we put a Q in the class $K(c_\lambda)$ equal to $\xi_\lambda^{c_\lambda} Q'$, then the Q' also form a system of power products none of which is divisible by another, and they contain only $n - 1$ variables. For $n = 2$ the Q' contain only one variable; their number is then 1, hence finite, so that the number may also be assumed finite for $n - 1$ variables. This proves the assertion.

Second, let $\mathfrak{M}$ be a module of polynomials whose terms are lexicographically ordered. Let $\mathfrak{P}$ be the system of the power products which occur in the highest term, each written only once. Then to the system $\mathfrak{Q} = (Q_1, \dots, Q_\rho)$ there corresponds a system of finitely many polynomials $F_1, \dots, F_\rho$, obtained by assigning

³Continuing this argument shows that among $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ every two are relatively prime; the converse, that this would imply $\mathfrak{M}_1, \dots, \mathfrak{M}_k$ to form a totally relatively prime system, is generally valid only in the commutative case; cf. the counterexample 1 in § 12.

to each Q_i some polynomial from $\mathfrak{M}$ whose highest term is equal to $b_i Q_i$, so that $F_i = b_i Q_i + \dots$ with lower terms.

Now let $F = bP + \dots$ be an arbitrary polynomial from $\mathfrak{M}$, where $P = \xi_1^{\beta_1} \dots \xi_n^{\beta_n} Q_i$. Form the product

$$\xi_1^{\beta_1} \dots \xi_n^{\beta_n} F_i = \xi_1^{\beta_1} \dots \xi_n^{\beta_n} (b_i Q_i + \dots) = c \xi_1^{\beta_1} \dots \xi_n^{\beta_n} Q_i + \dots = cP + \dots$$

Here $c \neq 0$ by (6), and all following terms are lower terms in the lexicographic order, since multiplication is commutative as regards the exponents, up to lower terms.⁴ The difference

$$F - \frac{b}{c} \xi_1^{\beta_1} \dots \xi_n^{\beta_n} F_i$$

then contains only lower terms and again belongs to $\mathfrak{M}$. The procedure therefore terminates after finitely many steps, so that $F_1, \dots, F_\rho$ are recognized as a module basis.

Theorem III. *Every system $\mathfrak{S}$ of infinitely many polynomials has a finite module basis $G_1, \dots, G_k$, so that every polynomial G from $\mathfrak{S}$ admits a representation $G = A_1 G_1 + \dots + A_k G_k$.*

The following is immediate.

Corollary. *Every system $\mathfrak{S}$ of infinitely many residue classes mod $\mathfrak{M}$ has a basis $\mathfrak{r}_1, \dots, \mathfrak{r}_k$, so that every $\mathfrak{r}$ from $\mathfrak{S}$ is representable in the form $\mathfrak{r} = A_1 \mathfrak{r}_1 + \dots + A_k \mathfrak{r}_k$.*

Namely, assign to each residue class some representative and consider the system of these representatives together with all polynomials from $\mathfrak{M}$.

We now turn to the proof of the following theorem.

Theorem IV. *Every residue group can be represented as a sum of finitely many irreducible constituents; equivalently, by Theorem II, every module can be represented as the least common multiple of finitely many irreducible modules forming a totally relatively prime system.*

Suppose, to the contrary, that $\mathfrak{G}$ is a sum of infinitely many constituents $\mathfrak{U}_1, \mathfrak{U}_2, \dots$ (cf. § 5), and let $\mathfrak{a}_1, \mathfrak{a}_2, \dots$ be the corresponding units, all of which are nonzero by definition. We form the system $\mathfrak{S}$ of all these units; by the corollary to Theorem III it has a finite basis $\mathfrak{a}_{i_1}, \dots, \mathfrak{a}_{i_v}$ with $i_1 < \dots < i_v$. Let $\mu > i_v$. Then

$$\mathfrak{a}_\mu = \mathfrak{r}_1 \mathfrak{a}_{i_1} + \dots + \mathfrak{r}_v \mathfrak{a}_{i_v}.$$

This is a linear relation between units, which is excluded by the definition. Hence in every additive decomposition of a residue group an irreducible constituent occurs after finitely many steps; because of commutativity of the decomposition, we can place it at the beginning. Thus, without loss of generality, we may assume that the $\mathfrak{U}_i$ themselves are irreducible. But by the preceding argument only finitely many constituents $\mathfrak{U}_i$ can occur. The assertion is proved.

§ 7. A Case of Unique Decomposition.

Let

$$\mathfrak{G} = \mathfrak{U}_1 + \dots + \mathfrak{U}_k = \mathfrak{B}_1 + \dots + \mathfrak{B}_l$$

be a residue group whose constituents $\mathfrak{U}_1, \dots, \mathfrak{U}_k, \mathfrak{B}_1, \dots, \mathfrak{B}_l$ are irreducible. Then

$$\mathfrak{r}\mathfrak{e} = \mathfrak{r}\mathfrak{a}_1 + \dots + \mathfrak{r}\mathfrak{a}_k = \mathfrak{r}\mathfrak{b}_1 + \dots + \mathfrak{r}\mathfrak{b}_l, \quad (22)$$

where $\mathfrak{a}_\chi$ is the residue class of $\mathfrak{G}$ isomorphically assigned to the unit class of $\mathfrak{U}_\chi$, and $\mathfrak{b}_\lambda$ is the one assigned to the unit class of $\mathfrak{B}_\lambda$. Replacing $\mathfrak{r}$ by $\mathfrak{r}\mathfrak{b}_\lambda$, with λ a fixed index in the range $1, \dots, l$, gives

$$\mathfrak{r}\mathfrak{b}_\lambda = \mathfrak{r}\mathfrak{b}_\lambda \mathfrak{a}_1 + \dots + \mathfrak{r}\mathfrak{b}_\lambda \mathfrak{a}_k. \quad (23)$$

This representation of the group $\mathfrak{B}_\lambda$ is unique, but it is not an additive decomposition of $\mathfrak{B}_\lambda$, since the individual residue classes on the right need not belong to $\mathfrak{B}_\lambda$. Since $\mathfrak{b}_\lambda$ does not vanish, not every product $\mathfrak{b}_\lambda \mathfrak{a}_\chi$ can be zero. Without loss of generality, let

$$\mathfrak{b}_\lambda \mathfrak{a}_1 \neq 0, \dots, \mathfrak{b}_\lambda \mathfrak{a}_\rho \neq 0, \quad \mathfrak{b}_\lambda \mathfrak{a}_{\rho+1} = 0, \dots, \mathfrak{b}_\lambda \mathfrak{a}_k = 0.$$

⁴This shows that in place of law (6) one could also use the somewhat less restrictive law

$$\xi_i a = a_i \xi_i + a_{i,i+1} \xi_{i+1} + \dots + a_{i,n} \xi_n + b_i \quad (a_i \neq 0)$$

for some fixed numbering of the ξ .

Then

$$\mathfrak{r}\mathfrak{b}_\lambda = \mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_1 + \cdots + \mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\rho.$$

For a fixed χ in the range $1, \dots, \rho$, consider the totality of those residue classes $\mathfrak{r}\mathfrak{b}_\lambda$ which are nonzero and for which also $\mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. The two systems $\mathfrak{r}\mathfrak{b}_\lambda$ and $\mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are isomorphic. First, the assignment is one-to-one: from $\mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ follows, by hypothesis, $\mathfrak{r}\mathfrak{b}_\lambda = 0$, and from the latter, by uniqueness of representation (23), $\mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$. Furthermore, sums correspond to sums, and products with arbitrary polynomials to the corresponding products. Keeping χ fixed, the same argument can be made for every λ with $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. For each $\mathfrak{a}_\chi$ there must be at least one such $\mathfrak{b}_\lambda$, since $\mathfrak{a}_\chi = (\mathfrak{b}_1 + \cdots + \mathfrak{b}_l)\mathfrak{a}_\chi \neq 0$.

We first assume that for each χ there is only one λ , and for each λ only one χ , such that $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$.

This correspondence implies $k = l$; moreover the notation may be chosen so that $\mathfrak{b}_\chi\mathfrak{a}_\chi \neq 0$ and $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$. We then say that the products $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme. In this case, by (23), the unit is $\mathfrak{b}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi$, and since

$$\mathfrak{a}_\chi = \mathfrak{b}_1\mathfrak{a}_\chi + \cdots + \mathfrak{b}_l\mathfrak{a}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi = \mathfrak{b}_\chi, \quad (24)$$

one has, in general, $\mathfrak{U}_\chi = \mathfrak{B}_\chi$. Thus there is uniqueness of decomposition, and the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ form a diagonal scheme.

If, for example, the commutative law holds specifically for the residue classes $\mathfrak{a}_\chi\mathfrak{b}_\lambda$, then the preceding hypothesis is always fulfilled. For in (23) each constituent $\mathfrak{r}\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{r}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ is then contained in $\mathfrak{B}_\lambda$, so that (23) gives an additive decomposition of the group $\mathfrak{B}_\lambda$. Since $\mathfrak{B}_\lambda$ is irreducible, all but one of the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are zero. By (24), correspondingly, for fixed χ only one $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ is different from zero, since otherwise (24), by interchangeability of $\mathfrak{a}_\chi$ and $\mathfrak{b}_\lambda$, would yield an additive decomposition of $\mathfrak{U}_\chi$, which is supposed to be irreducible.⁵

The same conclusions hold in the case where the conditions $\mathfrak{b}_\lambda\mathfrak{a}_\lambda \neq 0$, $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$, $\chi = 1, \dots, k$, and likewise, for each $\mu = 1, \dots, l$, $\mathfrak{b}_\mu\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\mu\mathfrak{a}_\mu \neq 0$, $\chi = 1, \dots, \sigma$, are fulfilled not for all, but only for part of the $\mathfrak{b}_\lambda$ ($\lambda = 1, \dots, \sigma$). One finds $\mathfrak{U}_\chi = \mathfrak{B}_\chi$ for $\chi = 1, \dots, \sigma$; and if one puts

$$\mathfrak{U}_{\sigma+1} + \cdots + \mathfrak{U}_k = \mathfrak{U}, \quad \mathfrak{B}_{\sigma+1} + \cdots + \mathfrak{B}_l = \mathfrak{B},$$

then, also for the units $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{U}$ and $\mathfrak{B}$, the conditions $\mathfrak{b}\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\chi\mathfrak{a} = 0$ ($\chi = 1, \dots, \sigma$), and $\mathfrak{b}\mathfrak{a} \neq 0$ are fulfilled, whence $\mathfrak{U} = \mathfrak{B}$.

§ 8. Additive Decomposition of Completely Reducible Groups into Prime Groups. The Theorem on Isomorphism.

We now consider another case, in which uniqueness does not hold, but isomorphism of the different decompositions does. This is the generalization of the case considered by Loewy.

Definition. A module is called a *prime module* if it has no divisor other than itself and the unit module containing all polynomials. The residue group of a prime module is called a *prime group*.

Every prime group is, a fortiori, irreducible. There are also infinite prime groups (cf. the definition in § 11 and example § 12, 2).

Let the given residue group $\mathfrak{G}$ now be representable in two ways as a sum of finitely many prime groups $\mathfrak{U}_1, \dots, \mathfrak{U}_k$ and $\mathfrak{B}_1, \dots, \mathfrak{B}_l$. A residue group which admits at least one such representation as a sum of prime groups, and the corresponding module $\mathfrak{M}$, will be called completely reducible, following Loewy's terminology.

We consider the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ ($\chi = 1, \dots, k$, $\lambda = 1, \dots, l$), and show that either $\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$, or else $\mathfrak{r}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ runs through the whole group $\mathfrak{B}_\lambda$ as $\mathfrak{r}$ runs through all residue classes of $\mathfrak{G}$. Indeed, the module belonging to $\mathfrak{B}_\lambda$ is

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{M}_{\mathfrak{b}_\lambda};$$

it arises from $\mathfrak{M}$ by adjoining all polynomials of the residue class $\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l$. This module has the divisor

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l, \mathfrak{a}_\chi\mathfrak{b}_\lambda).$$

But since $\mathfrak{M}_{\mathfrak{b}_\lambda}$ is, by hypothesis, a prime module, the divisor is either equal to $\mathfrak{M}_{\mathfrak{b}_\lambda}$ or to the unit module. In the first case $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ belongs to $\mathfrak{M}_{\mathfrak{b}_\lambda}$; hence there is a relation

$$\mathfrak{a}_\chi\mathfrak{b}_\lambda = \mathfrak{r}(\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l),$$

⁵The same applies in the noncommutative two-sided case treated by Schmeidler, loc. cit., where the product of every residue class of one subgroup with every residue class of the other subgroup vanishes. Then, since $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_1 + \cdots + \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_l$ and $\mathfrak{b}_\lambda(\mathfrak{a}_\chi\mathfrak{b}_\mu) = 0$ for $\lambda \neq \mu$, one has $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_\lambda$, hence this class is contained in $\mathfrak{B}_\lambda$; by irreducibility of $\mathfrak{B}_\lambda$, it follows that for only one, and hence for exactly one, χ the product $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. In the same way one shows that for every $\mathfrak{a}_\chi$ there is one and only one $\mathfrak{b}_\lambda$ with $\mathfrak{a}_\chi\mathfrak{b}_\lambda \neq 0$, whence $k = l$. Thus the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme, proving the uniqueness established there.

which gives $\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$. In the second case there are two classes $\mathfrak{r}_0$ and $\mathfrak{r}_1$ such that

$$\mathfrak{r}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda + \mathfrak{r}_1 (\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{e} = \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l + \mathfrak{b}_\lambda,$$

and therefore, by uniqueness, $\mathfrak{r}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda = \mathfrak{b}_\lambda$. Thus $F\mathfrak{r}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda$, and hence also $\mathfrak{r}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda$, runs through the whole group $\mathfrak{B}_\lambda$ as F or $\mathfrak{r}$ runs through all polynomials or all residue classes of $\mathfrak{G}$. This proves the assertion.

If $\mathfrak{a}_\chi \mathfrak{b}_\lambda \neq 0$, then $\mathfrak{U}_\chi$ is also isomorphic to $\mathfrak{B}_\lambda$. For by § 7 the totality of all classes $\mathfrak{r}_\chi \mathfrak{a}_\chi \neq 0$ for which $\mathfrak{r}_\chi \mathfrak{a}_\chi \mathfrak{b}_\lambda \neq 0$ is isomorphic to the system of classes $\mathfrak{r}_\chi \mathfrak{a}_\chi \mathfrak{b}_\lambda$; and these exhaust the subgroup $\mathfrak{B}_\lambda$. It remains only to show that the indicated classes $\mathfrak{r}_\chi \mathfrak{a}_\chi$ exhaust $\mathfrak{U}_\chi$. To this end, consider all residue classes $\mathfrak{r}_\chi \mathfrak{a}_\chi$ for which $\mathfrak{r}_\chi \mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$. This system has the property that the sum of two of its classes and the product with an arbitrary polynomial again belong to the system. If all these residue classes are adjoined to the module $\mathfrak{M}_{\mathfrak{a}_\chi}$, one again obtains a module which is a divisor of $\mathfrak{M}_{\mathfrak{a}_\chi}$; thus it is either $\mathfrak{M}_{\mathfrak{a}_\chi}$ or the unit module. In the latter case, however, $\mathfrak{a}_\chi$ itself would be contained in it, so that $\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$, contrary to the hypothesis. Hence there are no classes $\mathfrak{r}_\chi \mathfrak{a}_\chi \neq 0$ for which $\mathfrak{r}_\chi \mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$, and the isomorphism of $\mathfrak{U}_\chi$ with $\mathfrak{B}_\lambda$ is proved.

Suppose now, for a fixed χ , that $\mathfrak{a}_\chi \mathfrak{b}_{\lambda_1} \neq 0, \dots, \mathfrak{a}_\chi \mathfrak{b}_{\lambda_i} \neq 0$ with $i > 1$. Then $\mathfrak{U}_\chi$ is isomorphic to the groups $\mathfrak{B}_{\lambda_1}, \dots, \mathfrak{B}_{\lambda_i}$, so these groups are mutually isomorphic.

We now show that at least two of the $\mathfrak{U}$ must then also be isomorphic. For if in the products $\mathfrak{b}_\lambda \mathfrak{a}_\chi$ there belonged to each $\mathfrak{b}_\lambda$ only a single $\mathfrak{a}_\chi$ with $\mathfrak{b}_\lambda \mathfrak{a}_\chi \neq 0$, so that $\mathfrak{b}_\lambda = \mathfrak{b}_\lambda \mathfrak{a}_\chi$, then $\mathfrak{B}_\lambda = \mathfrak{U}_\chi$, since $\mathfrak{U}_\chi$, as a prime group, cannot contain a proper subgroup. Thus $k = l$ and, with a correct numbering, $\mathfrak{a}_\chi = \mathfrak{b}_\chi$; but then the scheme $\mathfrak{a}_\chi \mathfrak{b}_\lambda$ would be a diagonal scheme, which is not the case. The following theorem is therefore proved.

Theorem V. *If a completely reducible group $\mathfrak{G}$ is represented in two ways as a sum of prime groups,*

$$\mathfrak{G} = \mathfrak{U}_1 + \cdots + \mathfrak{U}_k = \mathfrak{B}_1 + \cdots + \mathfrak{B}_l,$$

then every group $\mathfrak{U}$ is isomorphic to at least one group $\mathfrak{B}$, and conversely. If every $\mathfrak{U}$ is isomorphic to exactly one $\mathfrak{B}$, then the decompositions are identical. In the other case at least two groups $\mathfrak{U}$ are mutually isomorphic, and likewise at least two groups $\mathfrak{B}$ are mutually isomorphic.

§ 9. Connection Between the Concepts “Isomorphic” and “of the Same Kind”.

In this paragraph we show that the concept “of the same kind”, known for differential expressions in one variable, when generalized in the natural way, leads to the isomorphism of the associated residue groups.

Definition.⁶ *Two modules $\mathfrak{M}$ and $\mathfrak{N}$ are called of the same kind if there are two polynomials P and Q , with P relatively prime to $\mathfrak{M}$ and Q relatively prime to $\mathfrak{N}$, such that for every $M \equiv 0(\mathfrak{M})$ and $N \equiv 0(\mathfrak{N})$*

$$MQ \equiv 0(\mathfrak{N}), \tag{25}$$

$$NP \equiv 0(\mathfrak{M}) \tag{26}$$

hold. Relative primeness here means the existence of two polynomials X and Y such that

$$YQ \equiv 1(\mathfrak{N}), \tag{27}$$

$$XP \equiv 1(\mathfrak{M}) \tag{28}$$

hold.

We now have the following theorem.

Theorem VI. *Two modules are of the same kind if and only if their residue groups are isomorphic.*

1. Let $\mathfrak{A}$ be the residue group of $\mathfrak{M}$, $\mathfrak{B}$ the residue group of $\mathfrak{N}$, and suppose $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$. Let $\mathfrak{a}$ and $\mathfrak{b}$ be the unit classes of $\mathfrak{A}$ and $\mathfrak{B}$. Further, let $Q\mathfrak{b}$ be the class in $\mathfrak{B}$ corresponding to the unit $\mathfrak{a}$ in $\mathfrak{A}$, and let $P\mathfrak{a}$ be the class in $\mathfrak{A}$ corresponding to the unit $\mathfrak{b}$ of $\mathfrak{B}$. We write this assignment as

$$\mathfrak{a} \sim Q\mathfrak{b}, \tag{29}$$

$$P\mathfrak{a} \sim \mathfrak{b}. \tag{30}$$

It follows that

$$0 = M\mathfrak{a} \sim MQ\mathfrak{b}, \quad \text{that is,} \quad MQ \equiv 0(\mathfrak{N}),$$

which proves (25). Likewise,

$$NP\mathfrak{a} \sim N\mathfrak{b} = 0, \quad \text{that is,} \quad NP \equiv 0(\mathfrak{M}),$$

⁶For differential expressions in one variable, cf. Blumberg's formal definition, loc. cit., p. 25.

which proves (26). Moreover, by (29), $P\mathfrak{a} \sim PQ\mathfrak{b}$; since by (30) $P\mathfrak{a} \sim \mathfrak{b}$ was assumed, uniqueness of the assignment gives $PQ\mathfrak{b} = \mathfrak{b}$, hence $PQ \equiv 1(\mathfrak{M})$, and similarly $QP \equiv 1(\mathfrak{M})$. Thus (27) and (28) are proved, with $X = Q$, $Y = P$.

2. Conversely, suppose $\mathfrak{M}$ and $\mathfrak{N}$ are of the same kind, so that relations (25) through (28) hold. We assign to an arbitrary class $F\mathfrak{a}$ of $\mathfrak{A}$ the class $FQ\mathfrak{b}$ of $\mathfrak{B}$. This assigns to each class in $\mathfrak{A}$ only one class in $\mathfrak{B}$: for from $F\mathfrak{a} = 0$, hence $F = M$, it follows by (25) that $FQ\mathfrak{b} = 0$. Moreover, the whole group $\mathfrak{B}$ is exhausted by this assignment, since the special class $Y\mathfrak{a}$ corresponds to $YQ\mathfrak{b}$, which equals $\mathfrak{b}$ by (27). Thus we have an unambiguous map Φ from $\mathfrak{A}$ onto $\mathfrak{B}$; it has not yet been shown to be one-to-one. Similarly, equations (26) and (28) give an unambiguous map Ψ from $\mathfrak{B}$ onto $\mathfrak{A}$, which again has not yet been shown to be one-to-one.

If either of these two maps is one-to-one, then the isomorphism is proved, since the requirements concerning sums and products are fulfilled. But if both maps Φ and Ψ were not one-to-one, then there would be classes $G\mathfrak{a}$ in $\mathfrak{A}$ corresponding to the class $GQ\mathfrak{b} = 0$ in $\mathfrak{B}$. The classes $FQ\mathfrak{b}$ corresponding to all the other residue classes $F\mathfrak{a}$ would then already exhaust the group $\mathfrak{B}$, and consequently the map Ψ would return the whole group $\mathfrak{A}$ in the classes $FQPa$. As we know, there are also nonzero classes in $\mathfrak{B}$ which, under Ψ , correspond to the zero class in $\mathfrak{A}$; we denote by $G_1\mathfrak{a}$ all those classes for which $G_1\mathfrak{a} \neq 0$ but $G_1QPa = 0$. All remaining polynomials F have the property that $FQPa$ exhausts the group $\mathfrak{A}$. Those polynomials F for which $FQPa = G_1\mathfrak{a}$ will be denoted by G_2 ; such polynomials exist whenever polynomials G_1 exist. For each G_2 one has $G_2QPa \neq 0$, but $G_2(QP)^2\mathfrak{a} = 0$. Correspondingly, for every integer ν there are classes $G_\nu\mathfrak{a}$ such that $G_\nu(QP)^{\nu-1}\mathfrak{a} \neq 0$ but $G_\nu(QP)^\nu\mathfrak{a} = 0$.

Now consider the system $\mathfrak{S}$ consisting of all polynomials $G_1, G_2, \dots$. It has a finite module basis $G_{\lambda_1}, \dots, G_{\lambda_e}$. Choose ν greater than the largest of the indices $\lambda_1, \dots, \lambda_e$, which we denote by λ . Then

$$G_\nu = A_1 G_{\lambda_1} + \dots + A_e G_{\lambda_e}.$$

Multiplying by $(QP)^\lambda$ gives

$$G_\nu(QP)^\lambda \equiv 0(\mathfrak{M}),$$

a contradiction to our hypothesis. Hence the isomorphism follows, and therefore, by part 1, the sharper relations (27), (28) hold with $X = Q$, $Y = P$.

By Theorem VI, Theorem V may also be expressed as follows.

Theorem VII.⁷ *If a completely reducible module $\mathfrak{M}$ is representable in two ways as the least common multiple of prime modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ and $\mathfrak{D}_1, \dots, \mathfrak{D}_l$, each forming a totally relatively prime system, then every $\mathfrak{P}$ is of the same kind as at least one $\mathfrak{D}$, and conversely. If every $\mathfrak{P}$ is of the same kind as exactly one $\mathfrak{D}$, then the decompositions are identical. In the other case at least two of the modules $\mathfrak{P}$ are of the same kind as one another, and likewise at least two of the modules $\mathfrak{D}$ are of the same kind as one another.*

⁷Cf. Loewy, pp. 100–101.